

10 | OVERVIEW


Climate Resilience for Frontline Clinics and the Patients They Serve

Abstract

Heat waves, wildfires, hurricanes, and floods are disrupting health care and harming patients. Harvard Chan C-CHANGE works with frontline clinics, which serve highly affected communities, to create practical tools that keep patients safe. This lesson explores how researchers used surveys, pilot testing, and feedback to build a climate resilience toolkit, heat alerts, and action plans for clinics across the United States.

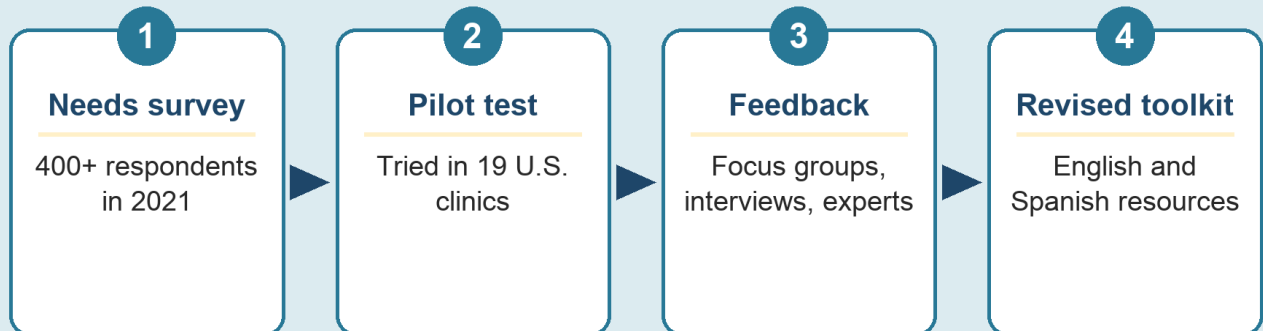
Introduction

Frontline clinics, such as free and charitable clinics and community health centers, often serve people who face the greatest risks from climate change. Nurses, doctors, and administrators need to know how to prepare their clinics and counsel their patients when extreme weather strikes. Yet many practicing clinicians never learned about climate change and health during their training. To fill this gap, Harvard Chan C-CHANGE partnered with the nonprofit AmeriCares to develop resources that combine the best available scientific evidence with the lived experience of health professionals. Their goal is simple: help clinics stay open and patients stay healthy when climate shocks arrive.

How Do Scientists Study This?

The project began with a needs assessment survey in 2021 that reached over 400 respondents. It showed that clinics needed information on climate hazards such as heat, wildfires, and extreme weather. The team then built the Climate Resilience for Frontline Clinics Toolkit and piloted an initial version in 19 clinics across the United States. Next, they gathered feedback through focus groups, surveys, interviews with toolkit users, and expert working groups. They used this feedback to revise the toolkit so it better met users' needs and reflected the latest research. This cycle of assessing needs, testing, listening, and improving is how researchers turn science into tools that work in the real world.

How the clinic toolkit was built



Assess needs

A 2021 survey of over 400 respondents identified the climate hazards clinics most needed help with.

Pilot the toolkit

An early version was tested in 19 clinics across the United States.

Listen and revise

Focus groups, surveys, interviews, and experts guided updates to meet users' needs.

What Did They Discover?

The resulting toolkit offers materials in English and Spanish for patients, providers, and clinic administrators, including patient action plans for heat, flooding, wildfires, and hurricanes. The team also created a heat alert system with Climate Central. It sends clinicians location-specific warnings from the CDC and National Weather Service HeatRisk system, along with information on who is at risk and how to counsel patients. With AmeriCares, the team built an online tool that clinics use to generate custom heat and wildfire smoke action plans, with steps ranging from wellness checks to air conditioner distribution. With support from Johnson & Johnson, the program now works with nearly 150 clinics and health centers.

Clinic resilience by the numbers

400+

respondents to the 2021 needs survey

19

clinics piloted the first toolkit

~150

clinics and health centers now in the program

The work is also expanding abroad: the team is adapting the toolkit for clinicians and patients in the Philippines with local health partners.

FROM THE ARTICLE



The Climate Resilience for Frontline Clinics Toolkit, developed with AmeriCares. Image: Harvard Chan C-CHANGE.



Stakeholder engagement workshop with the Department of Health of the Philippines, AmeriCares, and civil society, medical, and nursing groups, November 2023.

10 | TAKEAWAYS

Takeaways

Clinics are on the front line

Climate shocks disrupt care, so clinics need plans to protect patients.

Build with users

Surveys, pilots, and feedback shaped a toolkit that clinicians actually use.

Act before disaster

Heat alerts and action plans help clinics protect patients before harm occurs.

Key Terms

Frontline clinic

A health clinic, such as a free clinic or community health center, that serves communities most affected by health and climate risks.

Needs assessment

A survey or study that identifies what a group needs before designing a program or tool.

Pilot

A small first test of a program or tool to find out what works and what needs improvement.

HeatRisk

A CDC and National Weather Service system that forecasts how dangerous heat will be for health in a specific location.

Action plan

A written set of steps a clinic or patient follows to stay safe during a hazard such as heat or wildfire smoke.

10 | ACTIVITY

Activity

Clinic Heat Plan

Students create a one-page heat action plan for a community clinic and a matching tip sheet for patients.

Materials: Sample HeatRisk forecast map; Plan template; Markers or a slide tool

1. Identify three groups of patients who may be most at risk during extreme heat.
2. For clinic staff, list steps to take before, during, and after a heat alert, such as wellness checks.
3. Write a short patient tip sheet in plain language. Consider translating one tip into another language used in your community.
4. Swap plans with another group and give feedback, as focus groups did for the real toolkit.
5. Revise your plan based on the feedback you received.

References

Harvard T.H. Chan School of Public Health, Center for Climate, Health, and the Global Environment. Climate Resilience for Frontline Clinics and the Patients They Serve.

<https://hsph.harvard.edu/research/climate-health-c-change/outreach/building-climate-resilient-healthcare-clinics/>